



Earlier planting and late-maturing soybeans offset high temperature stress

Journal:	<i>Agricultural & Environmental Letters</i>
Manuscript ID	Draft
Wiley - Manuscript type:	Research Letter
Date Submitted by the Author:	n/a
Complete List of Authors:	Elli, Elvis; University of Arkansas dos Santos, Caio; Iowa State University of Science and Technology College of Agriculture and Life Sciences, Agronomy Fernandes, Samuel; University of Arkansas Salmerón, Montse ; University of Kentucky College of Agriculture Food and Environment Battisti, Rafael; Universidade Federal de Goiás De Souza Noia- Junior , Rogerio; LEPSE
Keywords:	phenology, warming, adaptive strategies, APSIM, elevated CO2

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Warmer summers can decrease U.S. soybean seed yields if agronomic management is not adapted accordingly. Our research question is: can farmers sustain soybean yields under high temperatures by changing variety maturity group and sowing dates? We quantified impacts of high temperatures on soybean yields across thousands of fields in the US Mid-South and investigated whether adapting sowing date and maturity group could offset negative impacts. Soybean yields dropped by nearly 6% on average with a +2°C temperature increase. However, planting earlier in the season and choosing a later-maturing soybean variety can increase seed yields by 9% on average under high temperatures. Maturity group selection and planting date are among the most actionable management decisions for farmers, as they can rapidly adopt these changed management practices without incurring significant production costs. These choices can be further optimized under warming to offset yield penalties.

**EARLIER PLANTING AND LATE-MATURING SOYBEANS OFFSET HIGH
TEMPERATURE STRESS**

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. We evaluated the APSIM model against observed regional data from the U.S. Mid-South and conducted gridded simulations across 4,651 fields to identify how concurrent changes in planting date and maturity group affect yield under rising temperatures and elevated [CO₂]. The +2°C temperature scenario reduced yields by 5.9% on average. However, combining a later-maturing soybean genotype with earlier sowing increased soybean yield by 9% under the +2°C scenario. These results revealed an opportunity to adapt soybean phenology through management and genotype selection to sustain yields under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shortening the crop cycle and limiting biomass accumulation and yield (Rose et al., 2016; Elli et al., 2022). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Zabel et al., 2021; Yang et al., 2025). Yet, it remains unclear how concurrent alterations in soybean maturity group and sowing date will affect yields, and whether existing genotypic availability and planting date flexibility are sufficient to offset the negative effects of high temperatures. This study advances

our understanding of how stacked agronomic management strategies influence soybean yield under elevated temperature in the U.S. Midsouth, a region particularly vulnerable to global warming.

Maturity group selection and planting date are among the most actionable management decisions for farmers to adapt to rising temperatures (Thornton et al., 2014). A relatively simple and well-established strategy to adjust soybean life-cycle to the available growing season length is to alter the soybean maturity group selection based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Earlier planting extends the duration of the growing season and shifts the crop cycle to cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While this agronomic practice has consistently increased soybean yield across the US, planting early is not always the best option, as there is an optimum window that maximizes capture of radiation resources and varies according to maturity group (Salmerón et al., 2016), indicating the need for cropping systems-specific recommendations.

Farmers have already been shifting towards earlier planting and selecting maturity groups well adapted to a region. However, there is limited information on whether these choices can be further optimized under warming and whether their interactive effects are sufficient to offset yield penalties. We hypothesize that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of accelerated developmental rates on soybean yield. We considered irrigated soybean production systems in the US Mid-South as a case study by conducting a regional-scale simulation at a fine spatial resolution ($\sim 0.025^\circ$; 4,651 fields) and across 40 years of weather data using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. Our objective was to

identify the roles of maturity group and sowing date in adapting soybean production to high-temperature risks under current and elevated [CO₂].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth comprising an 8.3° span in latitude, different planting dates, and cultivar maturity groups from 4.0 to 5.8, as described in Salmerón et al. (2016). The APSIM-Soybean model has previously proven successful for simulating the growth of various soybean maturity groups across water managements, soil types, and weather conditions in the US, including environments in the Mid-South (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data from Salmerón et al. (2016) and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018) version 2025.3.7681. Cultivar coefficients were calibrated for early MG 4 (relative MG - rMG 4.0 to 4.4) and early MG 5 (rMG from 5.0 to 5.4). The calibration results confirmed that the model is robust across a wide range of environmental conditions (Figures S1 and S2, and Table S1 in the supplementary material).

2.2. Regional simulations and scenario analysis

We conducted a simulation exercise across 40 weather years and 4,651 fields in the US Mid-South covering the delta region of the state of Arkansas, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2023). Further information on the weather conditions of the study area is presented in the supplementary material (Figure S3). We used the soybean

cropland data layer (NASS, 2023) to define the simulation fields, which were resampled to a 2.5 km grid ($\sim 0.025^\circ$) and clipped to the AR state boundary. The layer was reprojected to WGS84 to obtain latitude–longitude coordinates. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the National Weather Service Network using the *apsimx* R package. Soil water and inorganic nitrogen initial conditions were reset on January 1 each year and were held constant across locations to avoid confounding effects of soil vs. weather (Elli and Archontoulis, 2023). On the first day of the simulation (January 1), soil water content was set at field capacity, total inorganic nitrogen over the soil profile was 50 kg N ha^{-1} , and maize residue on the soil surface was 2000 kg ha^{-1} with a C:N ratio of 70. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile available.

For baseline simulations, the sowing date was May 15, an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2020 (NASS, 2023). We used an early 4 maturity group cultivar, which is well-adapted for the region (Salmeron et al., 2014; Salmerón et al., 2016). Soybean was irrigated as it is the predominant system across the studied region (NASS, 2023). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0–60 cm depth, to refill soil moisture to field capacity.

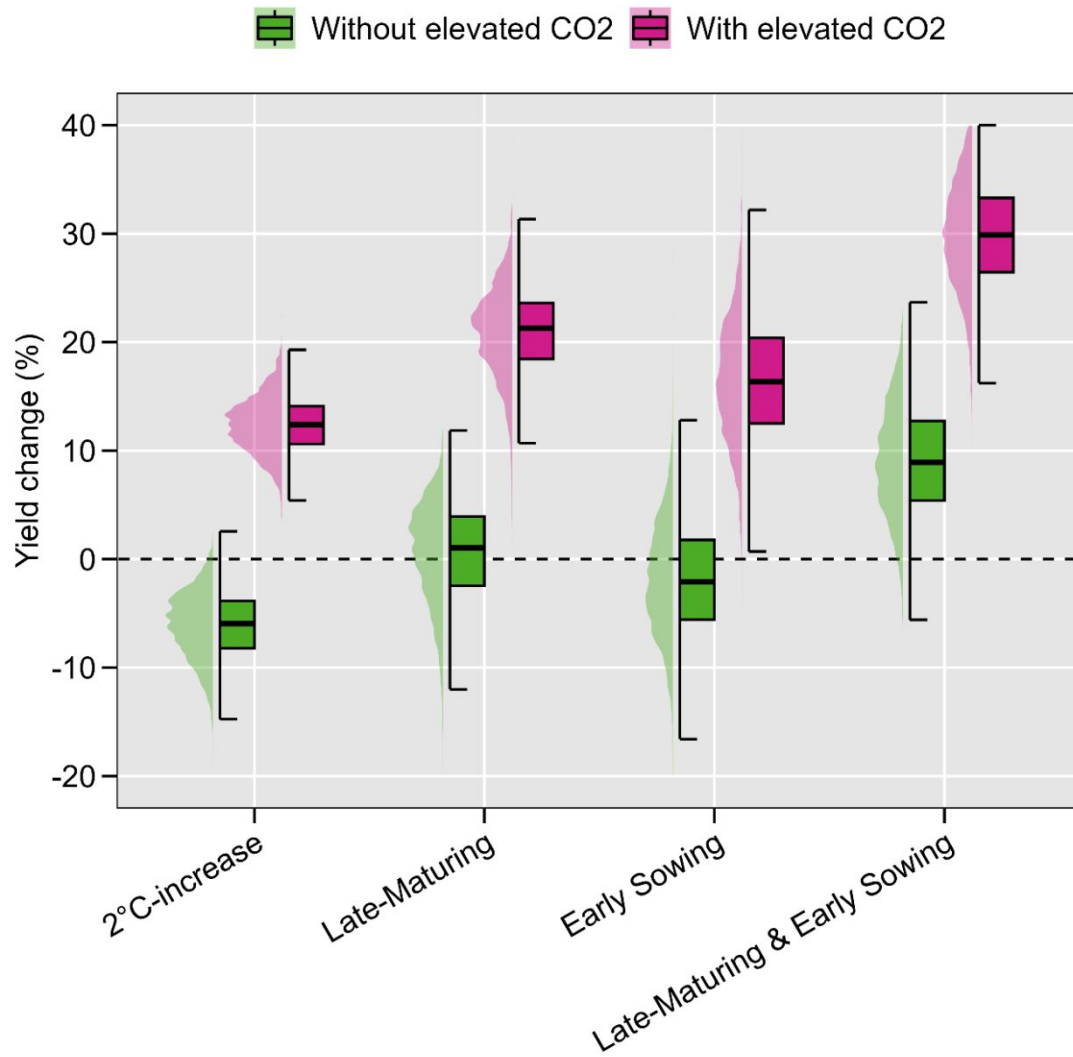
For the high-temperature scenario, we considered a 2°C increase in daily maximum and minimum temperatures, which is the average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A 2°C temperature increase combined with elevated $[\text{CO}_2]$ to 540 ppm was also evaluated, with the baseline set at 350 ppm (Elli and

Archontoulis, 2023). We employed a simpler approach rather than direct outputs from global circulation models because it has been proven to yield results with similar direction and magnitude, while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) advancing sowing date by 12 days, corresponding to a realistic range within the upper tail of the NASS planting progress for the studied region (NASS, 2023), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024).

3 RESULTS

3.1. Elevated temperature and [CO₂] impacts on soybean yields without adaptation

The 2°C temperature increase scenario decreased soybean seed yields by 5.9% on average, with values ranging from -14.8% to +2.6% across all site-years (Figure 1). The median simulated yield for the baseline condition was 4,525 kg ha⁻¹ with a south-to-north geographical gradient ranging from 2,235 to 4,981 kg ha⁻¹ (Figure S4). Under the 2°C temperature increase scenario, soybean seed yield decreased by 117 kg ha⁻¹ for every degree increase in season temperature (May-October, Figure 2). Yield response to temperature depended on local site-year weather conditions, with a reduction of 167 kg ha⁻¹ in seed yield for every degree increase in temperature under warm and dry conditions (Figure 2). Elevated [CO₂] increased seed yields and offset the negative effects of higher temperatures, with yields 12.4% greater under combined elevated temperature and [CO₂] than under baseline conditions, and increases ranging from 5.4 to 19.3% across site-years (Figure 1).



129

130 **Figure 1.** Overall relative effects of rising temperatures and adaptive genotypic and agronomic
 131 management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years
 132 in the US Mid-South.

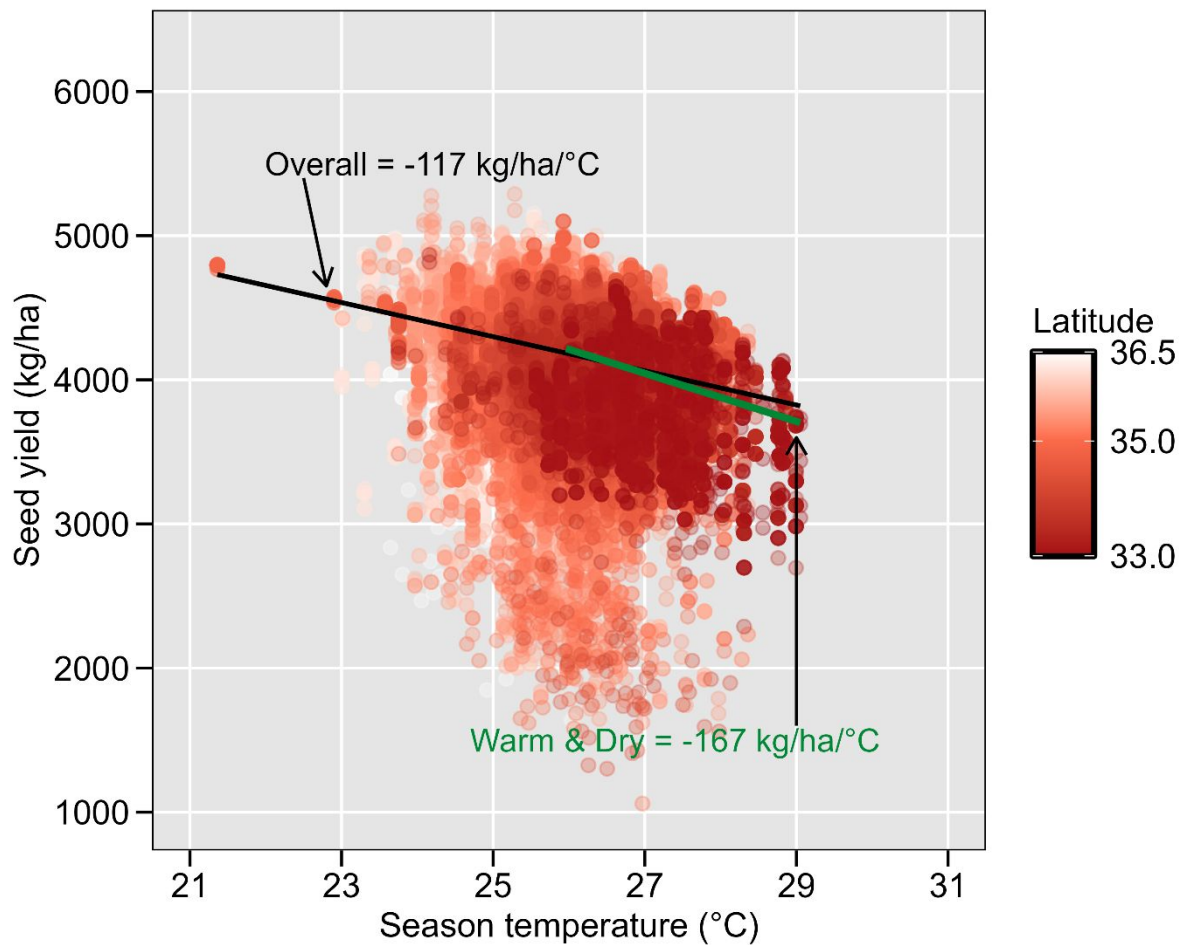


Figure 2. Relationship between season temperature (May-October) and soybean seed yield across 40 years and 4,651 fields in the US Mid-South under a 2 °C temperature increase scenario. The text in the panel represents the equation slopes for the overall data and for filtered Warm/Dry site-years.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone options to increase soybean yields under the high-temperature scenario (Figure 1). Using a late-maturing soybean genotype in combination with early sowing increased soybean seed yield by 9% under

the 2°C temperature increase scenario, without elevated CO₂. The standalone strategy of using a late-maturing cultivar under a 2°C temperature increase yielded median yields comparable to the historical baseline. This practice increased yields in 61% of the 4,651 fields simulated, particularly those located in the central-north region (Figure S5). Early sowing increased yields in 22% of the fields, mostly located in the Northeast region, compared to the 2°C temperature increase scenario. Elevated [CO₂] altered the magnitude but not the direction of the yield response to stacked adaptive management options, with the combination of genetic and management strategies being the most promising regardless of the [CO₂]. The combined impact of elevated CO₂ and stacked agronomic management strategies increased yields by 30% on average, with values ranging from 16% to 44% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by adjusting soybean phenology through cultivar maturity and planting date shifts. Combined genotypic and agronomic strategies emerged as the most promising approach for designing more resilient soybean production systems under a 2°C scenario. Our study extends previous work by quantifying the synergies and trade-offs among management adjustments aimed at mitigating heat risks in irrigated soybean systems of the U.S. Mid-South. Adjusting soybean maturity group to compensate for altered development rates due to rising temperatures combined with early sowing is likely among the most actionable solutions, as farmers can rapidly adopt these changed management practices without incurring significant production costs.

Our study revealed a 5.9% decrease in soybean yield due to rising temperatures, with values ranging from -14.8% to +2.6% across site-years (Figure 1). Elli et al. (2022) reported a 19% average reduction in seed yield across 10 locations across the US Corn Belt under a 2°C

166 temperature increase scenario without adaptive strategies. Deryng et al. (2014) investigated the
167 effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080.
168 Nendel et al. (2023) projected that heat stress will increasingly outweigh drought stress as a
169 constraint to soybean production in Europe. In our study soybean yield decreased by 2.8% (117
170 kg ha⁻¹ per °C) for every degree increase in temperature, which is higher than the 1.3% yield
171 reduction estimated by Lobell and Field (2007) on a global scale.

172 While elevated CO₂ can offset or alleviate the negative impacts of rising temperatures, as also
173 indicated in experimental studies (Ainsworth et al., 2002), genotypic differences in CO₂ response
174 may exist (Bishop et al., 2015). Moreover, model uncertainties exist in predicting yield benefits
175 under elevated [CO₂] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-
176 tolerant crop, heat events above 28°C during the reproductive phase can negatively impact yield
177 by reducing seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The
178 APSIM-soybean version used in the present study accounts for temperature effects on seed
179 growth rate (Kothari et al., 2024), but does not account for the impacts of extreme heat and
180 drought on seed number determination, which is a common limitation in process-based models
181 (Richetti et al., 2025).

182 While warmer springs and longer frost-free periods have allowed farmers to plant earlier every
183 year (Mourtzinis et al., 2019), this practice is not always feasible due to wet springs and extreme
184 rainfall events, which are projected to intensify further (Elli et al., 2023). Given the important
185 role of adapting the soybean life cycle to changes in temperature, as demonstrated in this study,
186 further research is needed to enhance our understanding of soybean phenology and its genetic
187 and management control, to increase soybean seed yield under high temperature risks.

5 CONCLUSION

This study advances our understanding of standalone and stacked agronomic management options for sustaining soybean seed yields under high temperature risks. We found that soybean yields declined by 5.9% on average under rising temperatures. Adjusting soybean maturity group to compensate for altered development rates driven by rising temperatures, combined with 12-day early sowing, effectively prevented yield losses under high temperature. Standalone agronomic management strategies were less effective at overcoming the yield penalty than combined strategies but could be used in strategic environments.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmeron: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

ACKNOWLEDGMENTS

This study was supported by the University of Arkansas System Division of Agriculture (Elli and FFAR (Fernandes, project title: Integrating genomic prediction with crop modeling to develop climate-resilient cultivars). Sincere appreciation is extended to all the researchers responsible for collecting phenology data used for model testing.

SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4.0–5.8 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Table S1. Maturity group-specific default and optimized APSIM parameter values targeted in the optimization of early and late maturity groups 4 and 5 cultivars.

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5).

Figure S3. Environmental characterization of baseline conditions across all studied site-years (A) and number of calendar days to flower under baseline and the 2°C temperature increase scenario (B). Panel A contains the APSIM phenology response curve (0-1), where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

Figure S4. Density distribution in simulated soybean yields under baseline, climate change with and without CO₂ fertilization (A), and geographical distribution in soybean yields over the simulated study area under baseline, climate change (CC) with and without CO₂ fertilization (B).

Figure S5. Geographical distribution of the relative effects of rising temperatures on soybean seed yield and adaptive agronomic management strategies. Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing.

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Supplementary Material

EARLIER PLANTING AND LATE MATURING SOYBEANS OFFSET HIGH TEMPERATURE STRESS

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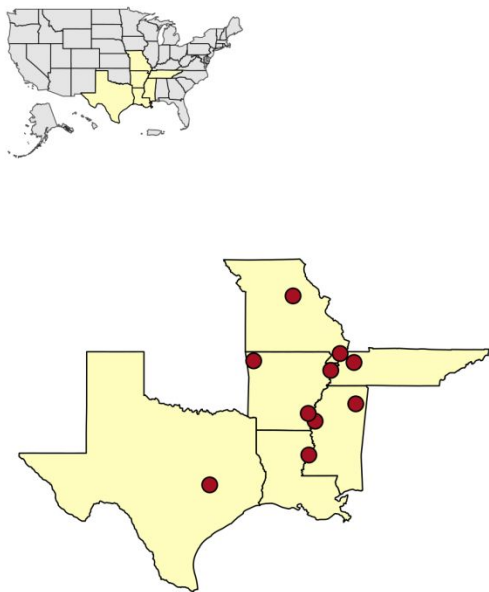
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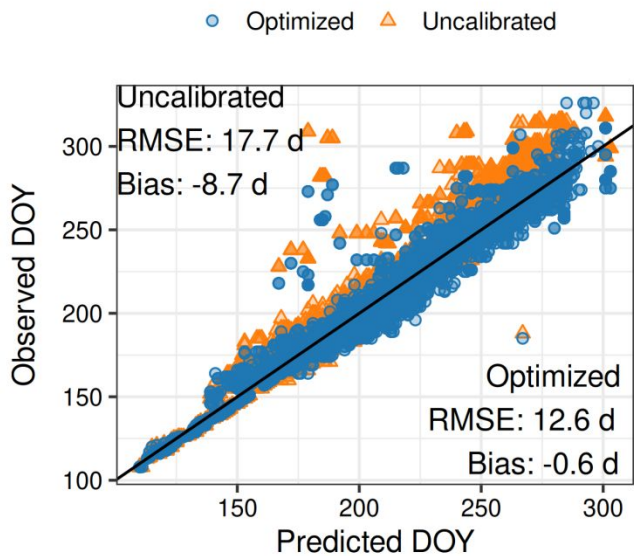
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A



B



C

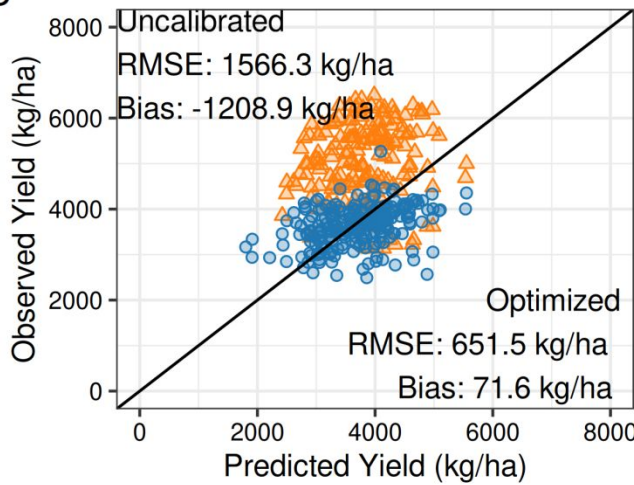


Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Model calibration

The occurrence of phenological stages, converted to days of the year (DOY), was used to derive the cultivar-specific model parameters (Table S1) using an iterative optimization procedure based on the Nelder–Mead algorithm (Nelder and Mead, 1965). This algorithm explores the parameter space iteratively using a geometric figure called simplex. The optimization procedure aims to minimize the residual sums of squares (Wallach et al., 2001) between the estimated and actual timing to reach the studied phenology stages for each maturity group (early 4, late 4, early 5, late 5). We ran the algorithm using R software (R Development Core Team, 2021) and the APSIMX package (Miguez, 2020). Our focus was on optimizing phenology parameters, while keeping cardinal temperature-related parameters constant, giving little evidence of changes across genotypes (Archontoulis et al., 2014). Two growth parameters were selected: area of the largest leaf and radiation use efficiency. The optimization workflow followed an approach similar to that outlined in Wallach et al. (2019), in which all parameters were optimized simultaneously to potentially capture parameter interactions.

Table S1. Maturity group (MG) – specific default and optimized APSIM parameter values targeted in the optimization of early and late maturity groups 4 and 5 cultivars.

APSIM Parameter	Units	Early MG 4	Early MG 5
Vegetative Target	°C-day	484	634
Early Flowering Target	°C-day	185	189
Early Grain Filling Target	Fraction (0-1)	0.338	0.042
Late Grain Filling Target	°C-day	384	418
Area of Largest Leaf	m ²	0.004	0.004
Radiation Use Efficiency	g MJ ⁻¹	1.48	1.05

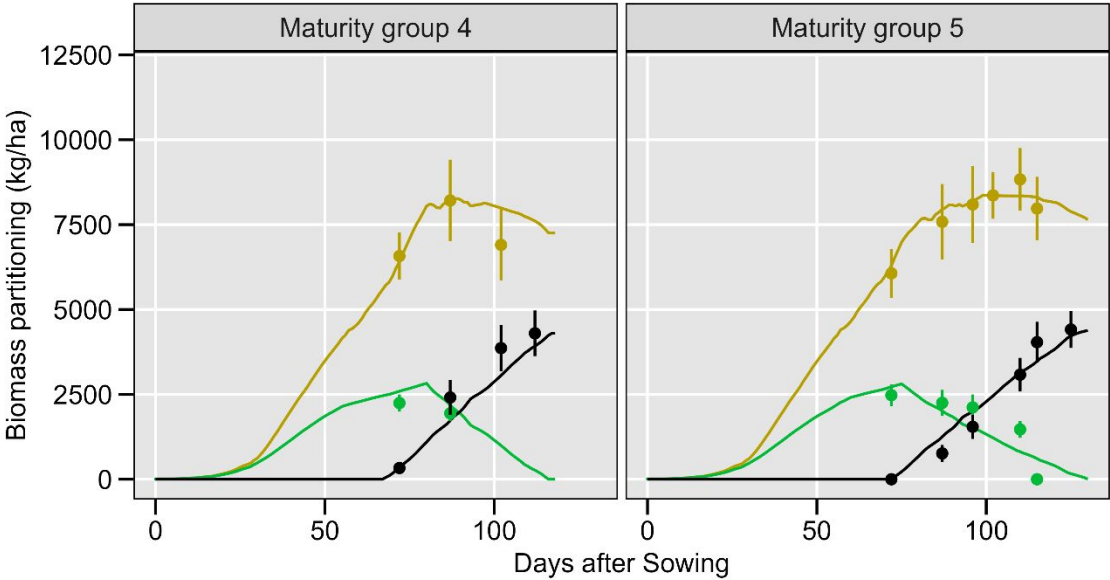


Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Study location and weather

The studied region is predominantly irrigated, with accumulated rainfall of 590 mm, unevenly distributed over the growing season (May to October). The average growing season temperature is 24°C, with values ranging from 20.9 to 27.1°C across all site-years (Figure S1). The studied area exhibits significant variability in soil conditions, ranging from shallow (less than 50 cm in depth) sandy soils with low soil water-holding capacity (0.4 mm cm^{-1}) to deeper soils with an effective root depth greater than 100 cm and high soil water-holding capacity (2 mm cm^{-1}). The soil organic carbon (0-30 cm) typically ranges from 0.5 to 3%.

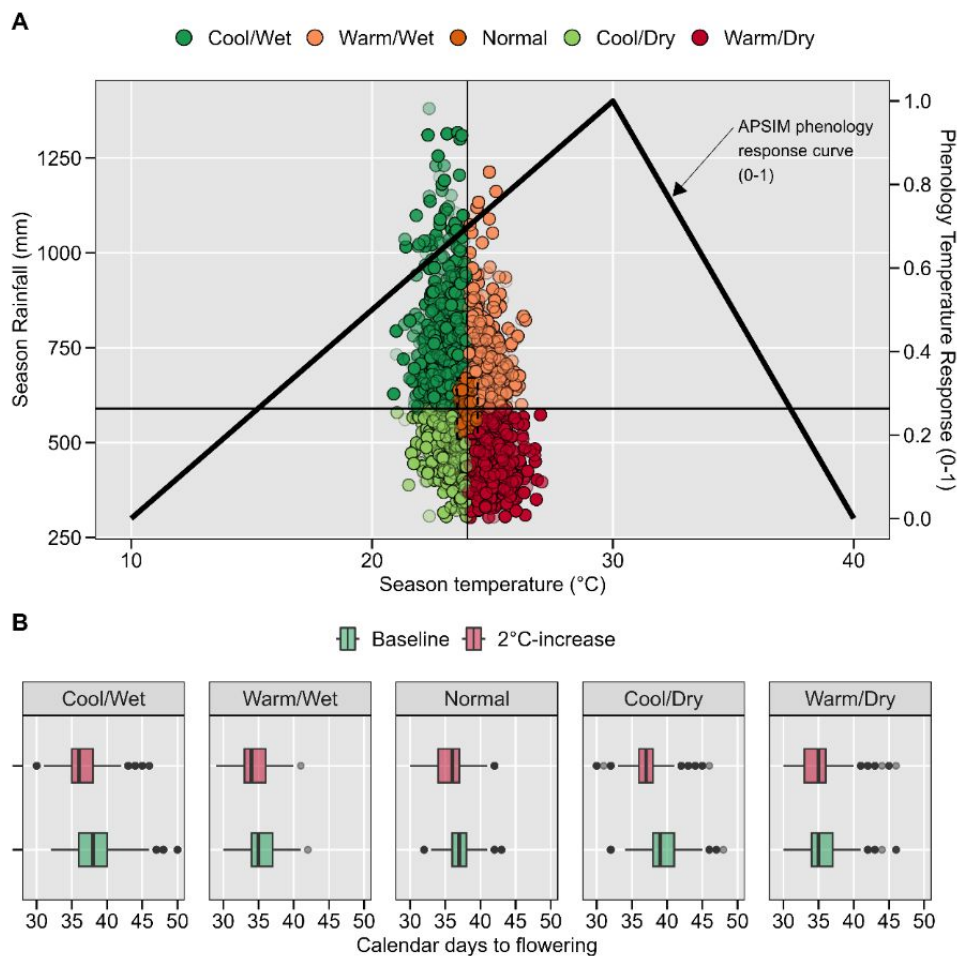


Figure S3. Environmental characterization of baseline conditions across all studied site-years (A) and number of calendar days to flower under baseline and the 2°C temperature increase scenario (B). Panel A contains the APSIM phenology response curve (0-1), where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

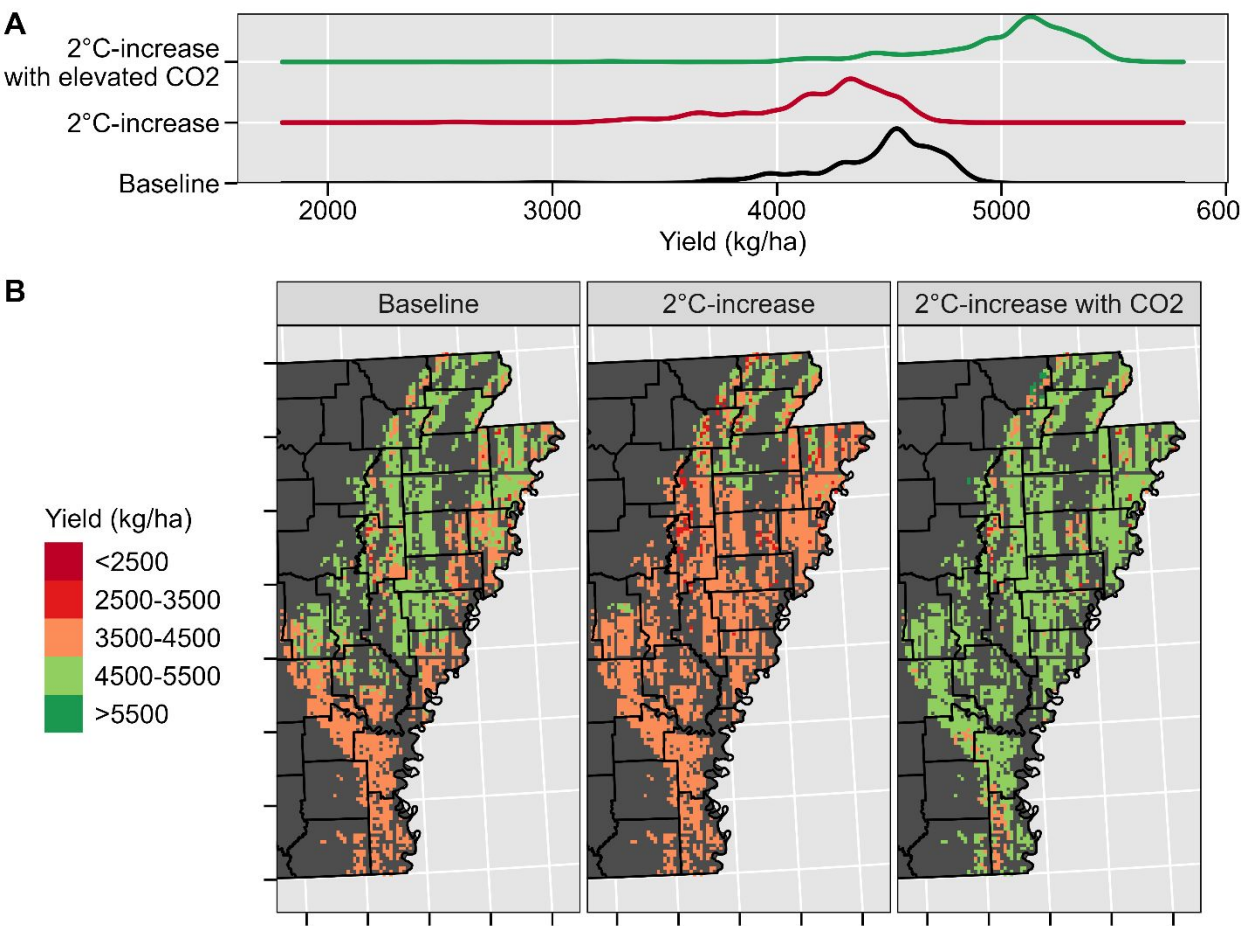


Figure S4. Density distribution in simulated soybean yields under baseline, climate change with and without CO₂ fertilization (A), and geographical distribution in soybean yields over the simulated study area under baseline, climate change (CC) with and without CO₂ fertilization (B).

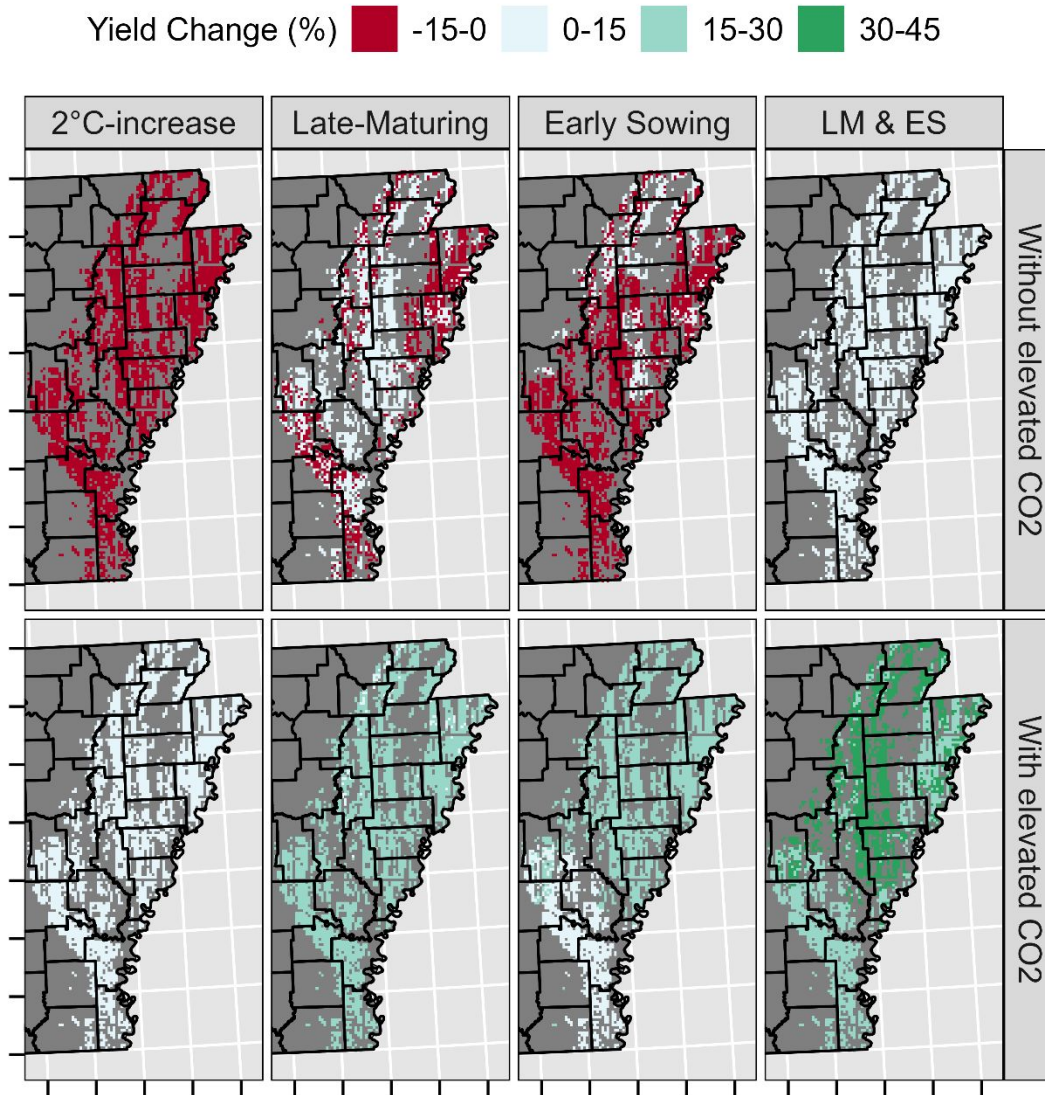


Figure S5. Geographical distribution of the relative effects of rising temperatures on soybean seed yield and adaptive agronomic management strategies. Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing.

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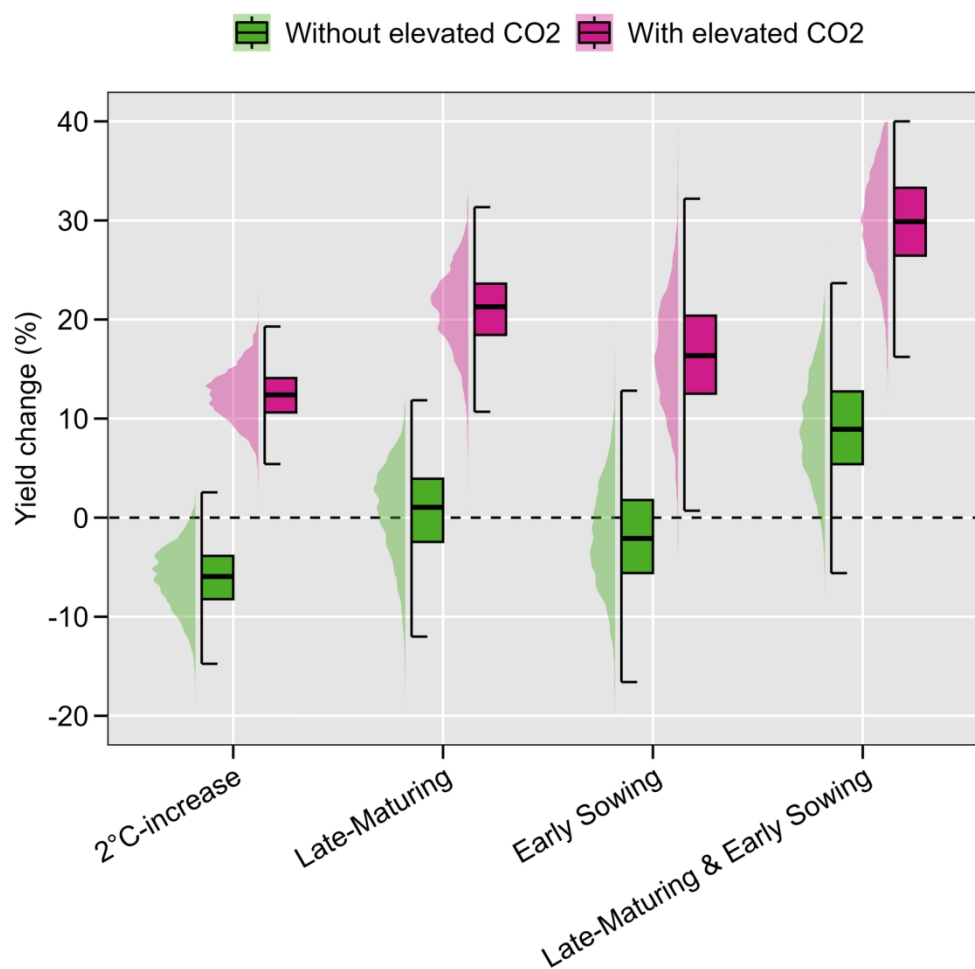


Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South.

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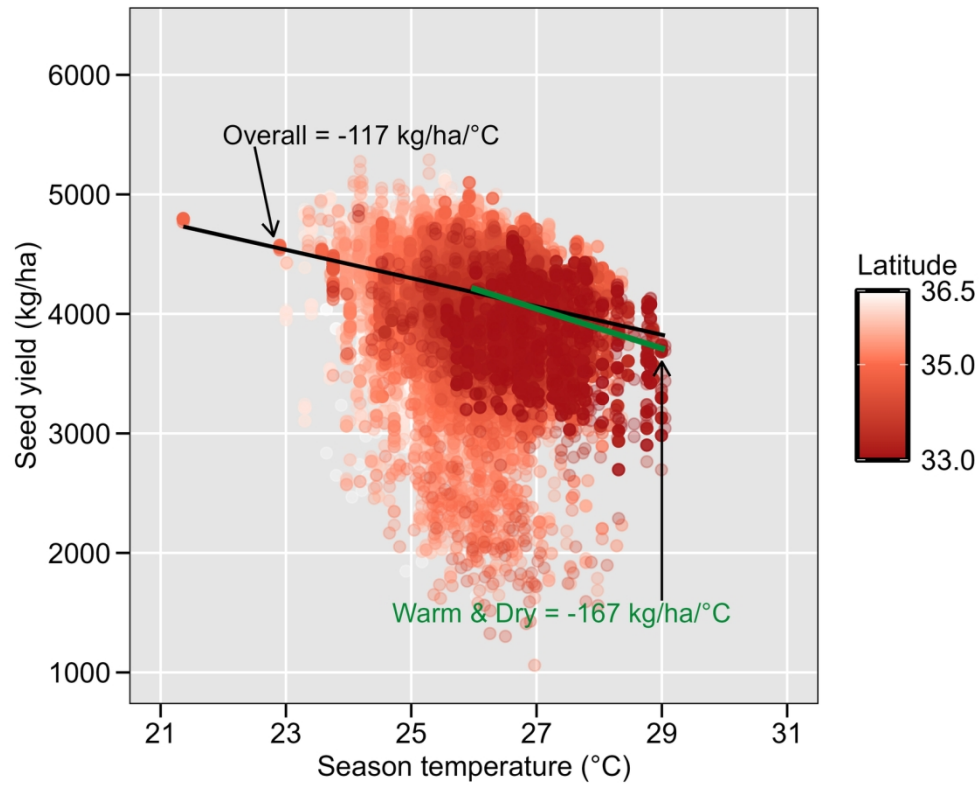


Figure 2. Relationship between season temperature (May-October) and soybean seed yield across 40 years and 4,651 fields in the US Mid-South under a 2 °C temperature increase scenario. The text in the panel represents the equation slopes for the overall data and for filtered Warm/Dry site-years.

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